AIRCRAFT FREIGHT

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

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2. Data Sources

- **Source Description and Timeline:** https://indiadataportal.com/p/airports-authority-ofindia/r/aai-airports_freight-ol-mn-aaa **and** 2017-2026.
- **Domain**: Aircraft Freight Analytics.

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3. Problem Statement

- To analyze aircraft freight data and understand overall freight performance.
- To clean and transform raw freight data using Microsoft Excel.
- To identify the top 5 states based on total freight volume.
- To identify the top 5 airports based on total freight volume.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Date	Date	Date of freight transportation.
State Code	Whole Number	Unique identification code of the state.
Airport Type	Text	Indicates whether the airport is Domestic or International.
Year	Whole Number	Numerical representation of the month (1–12).

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

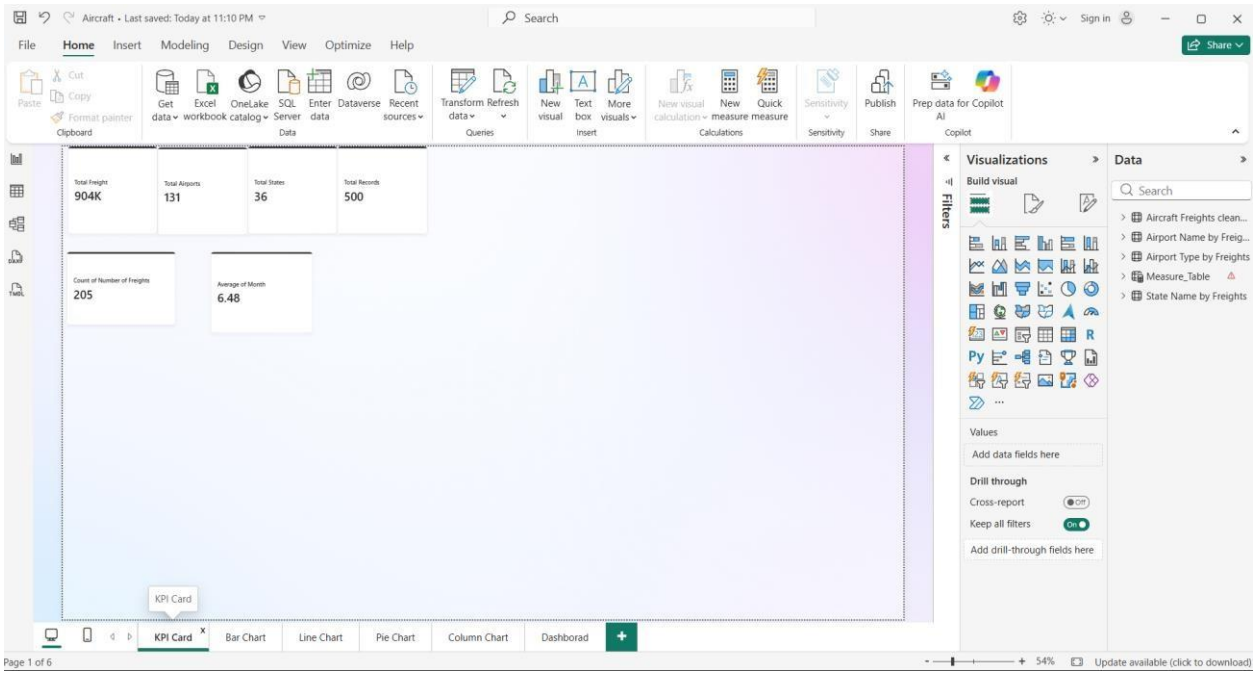
Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- Convert the data into Fact and Dimension Table **(If Possible)**

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7. Data Modelling and DAX (Power BI)

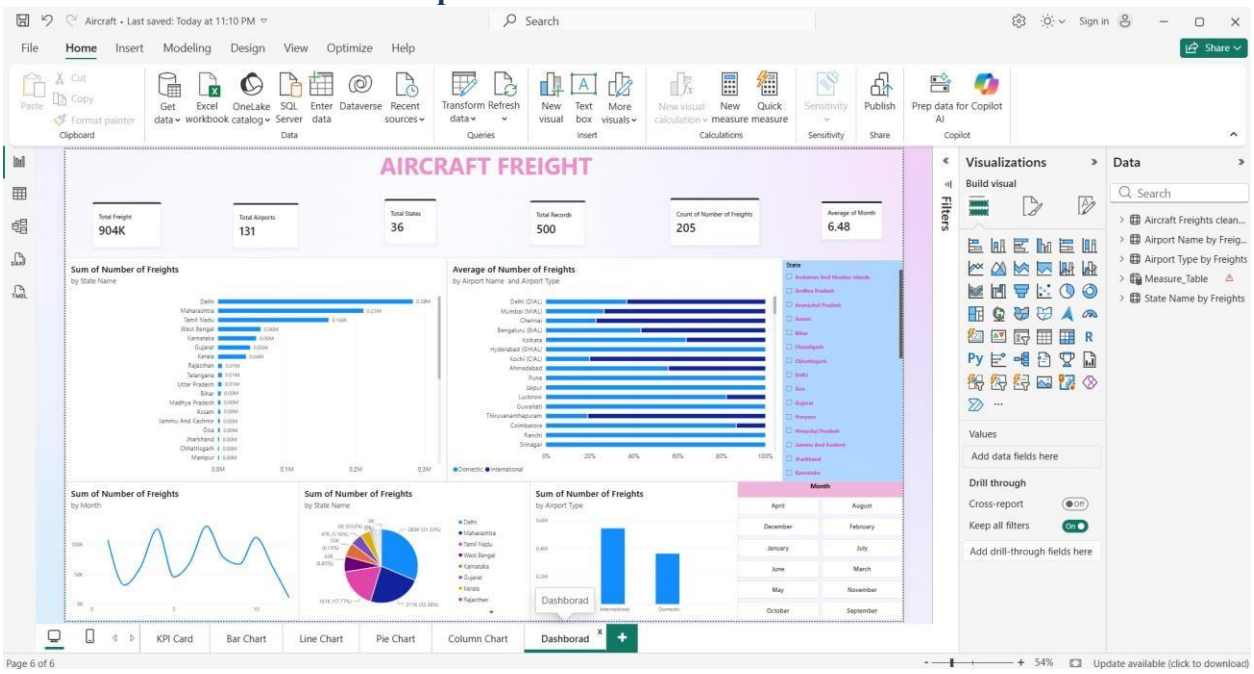
- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**.**



8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.
- Make your Report interactive Layout using **Drill-down**, **Filters and slicers • Bookmarks, Clear Titles for each visuals & Labels is must.**
- **Create a consolidated Report /Dashboard.**



9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data • **Provide the analysis insights: ▪ Descriptive**
  - ➢ Analyze the total number of freights handled by all airports.
  - ➢ Identify the top 5 states based on freight volume.
  - ➢ Ientify the top 5 airports based on freight volume.
  - ➢ Analyze monthly and quarterly freight trends.
  - ➢ Compare domestic and international airport freight volumes.

▪ **Diagnosis**

- Identify why some states have higher freight volumes than others.
- Analyze why certain airports handle more freight.
- Compare freight performance across different quarters.
- Identify states and airports with low freight volumes.
- Examine differences between domestic and international freight performance.

- **Predictive**
  - Predict upcoming monthly and quarterly freight demand.
  - Identify possible increases or decreases in freight volume.
  - Estimate future freight demand across different states.
  - Predict which airports may experience higher freight traffic.
- **Prescriptive**
  - Allocate resources to airports with higher freight demand.
  - Improve operational efficiency at low-performing airports.
  - Optimize freight handling and transportation operations.
  - Improve resource planning for future freight requirements.

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10. Conclusions

- The project analyzes aircraft freight data using Excel and Power BI to identify important trends and patterns.
- It helps compare state-wise and airport-wise freight performance and understand monthly and quarterly variations.
- The interactive dashboard supports better business decisions, resource planning and operational efficiency.